뇌는 역전파를 하지 않는다

2026. 06. 04.
ML 기초 시그 발표

전기정보공학부 25 방지현

Contents

역전파를 대체하려는 이유

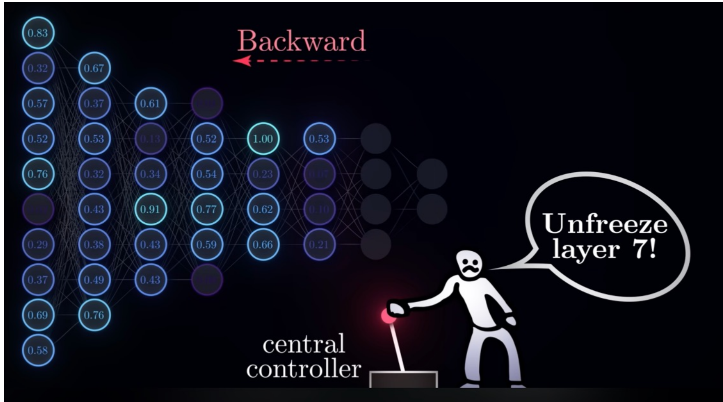
뇌는 역전파를 정말 안하는가? 근거

1. Weight Transport Problem

Gradient를 전달하려면 feedback과 feed forward 연결이 대칭이어야 함.
즉 **역방향 경로가 순방향 가중치를 알고 있어야** 함.

2. Two-phase problem

Feed forward phase인지 feedback phase인지 모든 셀이 알아야 함. - **지휘자?**
실제 **feedback이 올 때까지 freeze**한다면 우리는 그 동안 생각하지 못함.



뇌는 역전파를 정말 안하는가? 반론

1. Feedback Alignment

Backprop weight이 랜덤이어도 학습이 된다. (cf. 데이터 하나로도 학습되더라..)

=> 역방향 경로가 순방향 가중치를 몰라도 된다?

그러나 **Two-phase problem** 은 해결이 안된다.

결론: 뇌는 뉴런 단위의 역전파를 하지 않는다!

그렇다면 우리는 왜 역전파를 쓰는가?

1. 성능

역전파가 성능이 좋고, 현재까지 역전파를 이기는 대안이 없다.

2. 관성

Perceptron (1958) – Backpropagation (1986) – Deep learning (~2012)

공학적 최적화 도구로 자리 잡음. **굳이 생물학적 타당성을?**

요즘은 뇌보다 최적화를 잘하는 궁극적인 도구로 보는 시각도 있음.

그냥 역전파 그대로 쓰면 되지 않나?

“The human brain is the only existent proof we have,
perhaps in the entire universe,
that general intelligence is possible at all.”

- Demis Hassabis

+ 효율 (3.5MWh vs 50GWh)
20세 인간 GPT-4

어떻게 대체할건데?

뇌의 학습 메커니즘

Hebbian Learning (다음 페이지에)

Hebbian Learning and Engram

- **Donald Olding Hebb (1904-1985)**
- A Canadian neuropsychologist, postulated one of the most often quoted passage in Neuroscience
- Hebbian theory concerns how neurons might connected themselves to become engrams:

  'The general idea is an old one, that any two cells or systems that are repeatedly active at the same time will tend to become "associated" so that activity in one facilitates activity in the other'

  When one cell repeatedly assists in firing another, the axon of the first cell develops synaptic knobs (or enlarges them of they already exist) in contact with the soma of the second cell
- **Neurons that fire together, wire together**

뇌의 학습 메커니즘

Hebbian Learning

"Neurons that fire together, wire together" – Donald Hebb

⇒ Local Objective를 달성하고자 하는 것만으로도 지능이 창발된다??

(아직 밝혀진 게 많지 않음.. 미지의 영역)

어떤 시도들이 있었는가?

ML without backpropagation

연도	시도	저자	핵심 아이디어
1982	Hopfield Network	John Hopfield	에너지 최저점으로 수렴하며 학습
1985	Boltzmann Machine	Hinton & Sejnowski	Positive/Negative phase로 대조 학습
1989	Crick 비판	Francis Crick	"역전파는 뇌에서 일어날 수 없다"
1990년대	Contrastive Hebbian Learning	Movellan 등	Hebbian 규칙으로 오차 없이 학습
2014	Target Propagation	Yoshua Bengio	gradient 대신 target을 역방향으로 전달
2016	Feedback Alignment	Lillicrap 등	랜덤 역방향 가중치로도 학습 가능
2017	Equilibrium Propagation	Scellier & Bengio	에너지 고정점에서 gradient 계산, 역방향 패스 불필요
2022	Forward-Forward	Geoffrey Hinton	역방향 패스 제거, 두 번의 순방향으로 대체
2026	Hamiltonian Inference	Jack Kendall (Zyphra)	단 한 번의 순방향 패스로 수렴

에너지 기반

로컬 학습

Gradient 근사

V

V

The Forward-Forward Algorithm: Some Preliminary Investigations

Geoffrey Hinton
Google Brain
geoffhinton@google.com

Abstract

The aim of this paper is to introduce a new learning procedure for neural networks and to demonstrate that it works well enough on a few small problems to be worth further investigation. The Forward-Forward algorithm replaces the forward and backward passes of backpropagation by two forward passes, one with positive (*i.e.* real) data and the other with negative data which could be generated by the network itself. Each layer has its own objective function which is simply to have high goodness for positive data and low goodness for negative data. The sum of the squared activities in a layer can be used as the goodness but there are many other possibilities, including minus the sum of the squared activities. If the positive and negative passes could be separated in time, the negative passes could be done offline, which would make the learning much simpler in the positive pass and allow video to be pipelined through the network without ever storing activities or stopping to propagate derivatives.

1 What is wrong with backpropagation

The astonishing success of deep learning over the last decade has established the effectiveness of performing stochastic gradient descent with a large number of parameters and a lot of data. The gradients are usually computed using backpropagation (Rumelhart et al., 1986), and this has led to a lot of interest in whether the brain implements backpropagation or whether it has some other way of getting the gradients needed to adjust the weights on connections.

As a model of how cortex learns, backpropagation remains implausible despite considerable effort to invent ways in which it could be implemented by real neurons (Lillicrap et al., 2020; Richards and Lillicrap, 2019; Guerguiev et al., 2017; Scellier and Bengio, 2017). There is no convincing evidence that cortex explicitly propagates error derivatives or stores neural activities for use in a subsequent backward pass. The top-down connections from one cortical area to an area that is earlier in the visual pathway do not mirror the bottom-up connections as would be expected if backpropagation was being used in the visual system. Instead, they form loops in which neural activity goes through about half a dozen cortical layers in the two areas before arriving back where it started.

Forward-Forward Algorithm

2022, Geoffrey Hinton, 13 pages

Idea

Backprop: Forward pass - Feedback

FF Algorithm: Positive pass - Negative pass

ㄴ 각 레이어가 Goodness를 기준으로 독립 학습

활성도 제곱 합

Activation 계산 → Vector 길이(=Goodness) 측정

→ Weight update

→ 정규화 후 다음 레이어로 전달

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5 Sleep

FF would be much easier to implement in a brain if the positive data was processed when awake and the negative data was created by the network itself and processed during sleep.

In an early draft of this paper I reported that it was possible to do multiple updates on positive data followed by multiple updates on negative data, generated by the net itself, with very little loss in performance. The example I used was predicting the next character in a sequence from the previous ten character window. The use of discrete symbols rather than real-valued images makes it easier to generate sequences from the model. I have been unable to replicate this result and I now suspect it was due to a bug. Using the sum of squared activities as the goodness function, alternating between thousands of weight updates on positive data and thousands on negative data only works if the learning rate is very low and the momentum is extremely high. It is possible that a different goodness function would allow the positive and negative phases of learning to be separated, but this remains to be shown and it is probably the most important outstanding question about FF as a biological model.

If it is possible to separate the positive and negative phases it would be interesting to see if eliminating the negative phase updates for a while, leads to effects that mimic the devastating effects of severe sleep deprivation.

Forward-Forward Algorithm

2022, Geoffrey Hinton, 13 pages

성능 (error rate)

	역전파		FF
MNIST:	1.4%	vs	1.36%
CIFAR-10:	37%	vs	41%

논의 (Sleep)

Positive pass = 깨어있을 때,

Negative pass = 잘 때 => 버그였다고 함.

Equilibrium Propagation and Hamiltonian Inference in the Diffusive Fitzhugh-Nagumo Model

Jack Kendall*

Zyphra
San Francisco, CA
*Correspondence: jack@zyphra.com, tom@zyphra.com

***Abstract*—In this work, we extend the Equilibrium Propagation framework to skew-gradient systems and show an equivalence between deep Energy-Based Models and Hamiltonian neural networks. We focus on networks of diffusively coupled Fitzhugh-Nagumo neurons as a prototypical example. We show that since stationary solutions of the Fitzhugh-Nagumo model are described by self-adjoint operators, the methods of equilibrium propagation for performing credit assignment can be applied. Furthermore, for Fitzhugh-Nagumo networks with the topology of a deep residual network, we show that the steady state solutions admit a (spatial) Hamiltonian, and thus the methods of Hamiltonian Echo Backpropagation can be applied. We end by deriving an explicit layer-wise Hamiltonian recurrence relation governing inference for stationary solutions of both deep Fitzhugh-Nagumo networks and deep Energy-Based Models.**

I. Introduction

The question of how biological brains coordinate their synaptic updates across the brain in order to perform learning is a major open question in neuroscience, and is also relevant to machine learning. In machine learning, one typically uses stochastic gradient descent (SGD) to perform coordinated updates to all synapses with the goal of minimizing a loss or error function. SGD has the benefit of being both conceptually simple and highly effective in training deep or hierarchical neural networks, and is even able to train spiking and neuromorphic architectures (Lee et al., 2016; Wunderlich & Pehle, 2021).

However, in neuroscience, it is well-known that the backpropagation algorithm, which is used to estimate the gradients that SGD requires to coordinate its weight updates, is not biologically plausible (Grossberg, 1987). The primary reason for this is that backpropagation is a nonlocal algorithm: it requires an explicit "backward" graph along which to propagate gradient information, which must be matched to the forward graph at all times. This has led some neuroscientists to conclude that SGD itself is not biologically plausible as a learning algorithm. However, various alternatives to backpropagation

A. Related Work

One promising alternative to backpropagation for training biologically plausible neural networks is Equilibrium Propagation (EqProp) (Scellier & Bengio, 2017). EqProp applies to the class of networks known as Energy-Based Models (EBMs), such as Hopfield networks and their modern variants (Krotov & Hopfield, 2016).

Despite its restriction to EBMs, EqProp possesses several properties which make it attractive as a candidate for neural learning. Perhaps most importantly, it is fully local in its weight updates, i.e. the update for a given synapse in the network depends only on the local activity difference of the neurons to which it is connected. EqProp also possesses lower variance in the gradient estimate compared to node and weight perturbation and REINFORCE-like algorithms, which makes it scalable to neural networks of practical relevance (Høier et al., 2026; Scellier et al., 2023; Laborieux et al., 2021). At the same time, it yields low-bias gradient estimates compared to algorithms such as Feedback Alignment and Target Propagation (Laborieux et al., 2021). Lastly, the same network performs both inference and gradient estimation, so there is no need for a separate backward graph for propagating gradients.

Another approach, with a long history of use in neuroscience, is Predictive Coding (Friston, 2003; Bogacz, 2017; Millidge et al., 2021). EqProp and Predictive Coding are closely related, as both can be formulated in terms of a bilevel optimization problem: an inner optimization problem in which an energy function is minimized during inference, and an outer optimization problem where the weights are updated to perform learning. As a result, Predictive Coding and EqProp share many theoretical and empirical properties, such as equivalence to backpropagation under certain conditions (Millidge et al., 2022), and the fact that both algorithms optimize a well-defined objective function during learning. It is worth noting that Predictive Coding, unlike EqProp, is capable

Hamiltonian Inference

2026. 05. 20., Jack Kendall, 8 pages

Idea

Backprop: Euler's Method (경사하강)

Hamiltonian: 한번에 해 계산하기

(layer by layer, 점화식)

그게 가능하냐?

Parameter space: 너무 고차원이라 안됨.

Activation space: 평형 상태 계산

(Kirchhoff's law)

Hamiltonian이 레이어 방향으로 보존된다고 함..
공대 사람이라 수학을 잘 모르겠어요 ㅈㅅ

Equilibrium Propagation and Hamiltonian Inference in the Diffusive Fitzhugh-Nagumo Model

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Hamiltonian Inference

2026. 05. 20., Jack Kendall, 8 pages

성능

MNIST: error rate 2.8%

(cf. backprop 1.4% / FF 1.36%)

그러나..

30 layer 이상에서 성능 떨어짐.

그 전까지는 경사하강과 거의 동일한 결과!! 오..

결론: 뇌에서 배워야 한다!

아직 역전파를 대체할 정도는 아니지만, 시도가 많이 되고 있다!

전력 효율 문제가 커서 이제 정말 이런 방향의 돌파가 필요한 때이다.

질의응답

1. **사람은 뇌가 진화해왔기 때문에 에너지를 저걸로 치면 안되지 않냐?**
   ㄴ 그러니까 그 진화의 결과를 모방하려는 것.

2. **그러면 이미 Pretrain을 많이 해온 것도 어쩌면 진화의 결과 아닌가? 그냥 그러면 이미 있는 모델에서 발전시켜서 성능 잘 나오면 에너지 효율 면의 장점이 없어질 수 있는데 그래도 이런 시각이 경쟁력이 있는가?**
   ㄴ 나는 뇌과학을 하는 사람이기 때문에 뇌과학에 역으로 도움이 될 것 같아서 점수를 더 준다.
   also, benchmaxxing에 대한 생각. Backprop은 loss 정의를 기반으로 동작하기 때문에 다음 토큰 예측을 잘하게 학습하면 잘할 수 밖에 없음. 그러나 local optimization으로 다음 토큰 예측을 잘한다? 진짜 지능이라는 느낌이 좀 더 난다고 생각함.

\+ Perceptron이라는 책을 읽어볼 것.